Linear Algebra with Applications in Automatic Control

School of Automation,
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Fall Semester

4.4 Projector and idempotent matrices

Z. Song

- Projector-definition

Definition 4.10 Let $\mathcal{S}$, $\mathcal{T}$ be subspaces of $\mathcal{V}$ and $\mathcal{V} = \mathcal{S} \dotplus \mathcal{T}$, then any $\boldsymbol{\alpha}$ in $\mathcal{V}$ can be uniquely decomposed into the sum

$$\boldsymbol{\alpha} = \boldsymbol{\beta} + \boldsymbol{\gamma}, \quad \boldsymbol{\beta} \in \mathcal{S}, \quad \boldsymbol{\gamma} \in \mathcal{T} \tag{1}$$

$\boldsymbol{\beta}$ is called the projection of $\boldsymbol{\alpha}$ to $\mathcal{S}$ parallel to $\mathcal{T}$, and $\boldsymbol{\gamma}$ is called the projection of $\boldsymbol{\alpha}$ to $\mathcal{T}$ parallel to $\mathcal{S}$. The linear transformation

$$\mathscr{P}_{\mathcal{S},\mathcal{T}} : \mathcal{V} \mapsto \mathcal{S} \qquad \mathscr{P}_{\mathcal{S},\mathcal{T}}(\boldsymbol{\alpha}) = \boldsymbol{\beta}$$

determined by (1) is called the projector from $\mathcal{V}$ to $\mathcal{S}$ parallel to $\mathcal{T}$, while the linear transformation

$$\mathscr{P}_{\mathcal{T},\mathcal{S}} : \mathcal{V} \mapsto \mathcal{T} \qquad \mathscr{P}_{\mathcal{T},\mathcal{S}}(\boldsymbol{\alpha}) = \boldsymbol{y}$$

determined by (1) is called the projector from $\mathcal{V}$ to $\mathcal{T}$ parallel to $\mathcal{S}$.
When restricted on the subspace $\mathcal{S}$, $\mathscr{P}_{\mathcal{T},\mathcal{S}}$ is the identity transformation.

- See the intuitive explanation in 2-dimensional space on Pg 150

4.4 Projector and idempotent matrices

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- Projector-properties

Theorem 4.9 If $\mathscr{P}$ is a projector on $\mathcal{V}$, then

1. $\mathscr{E} - \mathscr{P}$ is a projector;

2. $\mathcal{R}(\mathscr{E} - \mathscr{P}) = \mathcal{N}(\mathscr{P})$;

3. $\mathcal{N}(\mathscr{E} - \mathscr{P}) = \mathcal{R}(\mathscr{P})$.

4.4 Projector and idempotent matrices

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- Projector-necessary and sufficient conditions

Theorem 4.10 $\mathscr{P} \in \mathcal{L}(\mathcal{V})$ is a projector from $\mathcal{V}$ to some subspace of $\mathcal{V}$ if and only if

$$\mathcal{V} = \mathcal{R}(\mathscr{P}) \dot{+} \mathcal{N}(\mathscr{P}).$$

Theorem 4.1 $\mathscr{P} \in \mathcal{L}(\mathcal{V})$ is a projector if and only if $\mathscr{P}^2 = \mathscr{P}$.

4.4 Projector and idempotent matrices

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- Projector-matrix representation- idempotent matrices

Definition 4.11 $A \in \mathcal{C}^{n\times n}$ is idempotent if

$$A^2 = A\,.$$

Theorem 4.12 The representation of a projector $\mathscr{P}$ on $\mathcal{V}$ is idempotent.

4.4 Projector and idempotent matrices

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- properties of idempotent matrices

Theorem 4.13 P is idempotent, then

1. P^{T}, P^*, $E-P$, $E-P^{\mathsf{T}}$, $E-P^*$ are all idempotent matrices;
2. $P(E-P)=(E-P)P=0$;
3. $\mathcal{N}(P)=\mathcal{R}(E-P)$;
4. $P\boldsymbol{x}=\boldsymbol{x}$ if and only if $\boldsymbol{x}\in\mathcal{R}(P)$;
5. $\mathcal{C}^n=\mathcal{R}(P)\dot{+}\mathcal{N}(P)$.

4.4 Projector and idempotent matrices

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- Properties of idempotent matrices

Theorem 4.14 P is an $n \times n$ idempotent matrix of rank r if and only if there exists $C \in \mathcal{C}^{n \times n}$ such that

$$C^{-1}PC = \begin{bmatrix} E_r & 0 \\ 0 & 0 \end{bmatrix} \qquad (1)$$

Corollary 4.3 $\text{rank}P = \text{tr}P$.

4.4 Projector and idempotent matrices

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- Orthogonal complement

Definition 4.12 Let $\mathcal{S}$ and $\mathcal{T}$ be subspaces of $\mathcal{V}$. If for all $\boldsymbol{\alpha} \in \mathcal{S}$, $\boldsymbol{\beta} \in \mathcal{T}$ the equality $\langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle = 0$ holds, then $\mathcal{S}$ and $\mathcal{T}$ are said to be mutually orthogonal, denoted $\mathcal{S} \perp \mathcal{T}$.

Theorem 4.15 Let $\mathcal{S}$ and $\mathcal{T}$ be mutually orthogonal subspaces of $\mathcal{V}$, then

1. $\mathcal{S} \cap \mathcal{T} = \mathbf{0}$, and

2. $\dim(\mathcal{S} + \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T}$

4.4 Projector and idempotent matrices

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- Orthogonal complement-continued

Definition 4.13 The sum of two mutually orthogonal subspaces $\mathcal{S}$ and $\mathcal{T}$ is called the orthogonal sum of $\mathcal{S}$ and $\mathcal{T}$, denoted $\mathcal{S} \oplus \mathcal{T}$. $\mathcal{S}$ is called the orthogonal complement of $\mathcal{T}$, denoted $\mathcal{S} = \mathcal{T}_\perp$, or

$$\mathcal{S} = \mathcal{T}_\perp = \{\boldsymbol{\alpha} \ : \ \langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle = 0, \quad \forall \boldsymbol{\beta} \in \mathcal{T}\}$$

Theorem 4.16 Let $\mathcal{T}$ be a subspace of $\mathcal{V}$. Then there exists a unique subspace $\mathcal{S} \subset \mathcal{V}$ such that

$$\mathcal{S} \oplus \mathcal{T} = \mathcal{V}$$

4.4 Projector and idempotent matrices

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•Orthogonal projectors

Definition 4.14 Suppose that $\mathcal{S} \oplus \mathcal{T} = \mathcal{V}$. If for any $\boldsymbol{\alpha} \in \mathcal{V}$, $\boldsymbol{\alpha} = \boldsymbol{\beta} + \boldsymbol{\gamma}$ with $\boldsymbol{\beta} \in \mathcal{S}$, $\boldsymbol{\gamma} \in \mathcal{T}$, the linear transformation $\mathscr{P} \in \mathcal{L}(\mathcal{V})$ is defined as

$$\mathscr{P}(\boldsymbol{\alpha}) = \boldsymbol{\beta}$$

Then $\mathscr{P}$ is referred to as the projector from $\mathcal{V}$ to the subspace $\mathcal{S}$.

Theorem 4.17 Let $\mathscr{P}$ be an orthogonal projector from $\mathcal{V}$ to $\mathcal{S}$. If $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_r$ are a orthonormal basis of $\mathcal{S}$, $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_r, \boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_n$ are an orthogonal basis of $\mathcal{V}$, then the representation of $\mathscr{P}$ with respect to the basis $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_r, \boldsymbol{u}_{r+1}, \ldots, \boldsymbol{u}_n$ is

$$P_{\mathcal{S}} = \begin{bmatrix} E_r & 0 \\ 0 & 0 \end{bmatrix}.$$

4.4 Projector and idempotent matrices

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- Orthogonal projectors-matrix representation properties

Definition 4.15 Let $\boldsymbol{a}_1, \boldsymbol{a}_2, \ldots, \boldsymbol{a}_n$ be an orthonormal system of n-dimensional column vectors, the $n \times r$ matrix $U_1 = [\boldsymbol{a}_1 \quad \boldsymbol{a}_2 \quad \ldots \quad \boldsymbol{a}_r]$ is called a sub-unitary matrix. The set of all $n \times r$ sub-unitary matrices are denoted $U_1 \in \mathcal{U}^{n\times r}$.

Theorem 4.18 $U_1 \in \mathcal{U}^{n\times r}$ if and only if $U_1^* U_1 = E_r$.

Theorem 4.19 Let A be $n \times n$ matrix. Then $A = A^* = A^2$ if and only if there exists $n \times r$ matrix $U \in \mathcal{U}^{n\times r}$ such that $A = UU^*$, where $r = \text{rank}A$.

4.5 Adjoint transformation

- Definition

Definition 4.16 Let $\mathcal{U}$ and $\mathcal{V}$ be n- and m-dimensional unitary spaces, $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$. $\mathscr{A}^* \in \mathcal{L}(\mathcal{V} \mapsto \mathcal{U})$ is referred to as the adjoint transformation of $\mathscr{A}$ if

$$\langle \mathscr{A}^*(\boldsymbol{\beta}), \boldsymbol{\alpha} \rangle_{\mathcal{U}} = \langle \boldsymbol{\beta}, \mathscr{A}(\boldsymbol{\alpha}) \rangle_{\mathcal{V}}, \quad \forall \boldsymbol{\alpha} \in \mathcal{U}, \quad \boldsymbol{\beta} \in \mathcal{V}.$$

- interpretation

Let $\mathcal{U}$ and $\mathcal{V}$ be n- and m-dimensional unitary spaces with the inner products $\langle \cdot, \cdot \rangle_{\mathcal{U}}$ and $\langle \cdot, \cdot \rangle_{\mathcal{V}}$ respectively, $\boldsymbol{\alpha}_1$, $\boldsymbol{\alpha}_2$, ..., $\boldsymbol{\alpha}_n$ and $\boldsymbol{\beta}_1$, $\boldsymbol{\beta}_2$, ..., $\boldsymbol{\beta}_m$ be the bases on $\mathcal{U}$ and $\mathcal{V}$. Let $\boldsymbol{\beta} \in \mathcal{V}$, with the representation $\boldsymbol{y} = [y_1 \quad y_2 \quad \dots \quad y_m]^{\mathsf{T}}$ relative to the basis of $\mathcal{V}$, then $\boldsymbol{\beta} = \sum_{i=1}^{m} y_i \boldsymbol{\beta}_i$.

4.5 Adjoint transformation

Denote by $G_{\mathcal{V}}$ the Gramian of the basis vectors in $\mathcal{V}$, then $G_{\mathcal{V}}$ is nonsingular and $\boldsymbol{y}$ can be uniquely determined by (4.1) (please refer to (4.1) on Pg 140.)

$$\boldsymbol{y} = G_{\mathcal{V}}^{-1}[\langle \boldsymbol{\beta}_1, \boldsymbol{\beta}\rangle_{\mathcal{V}} \quad \langle \boldsymbol{\beta}_2, \boldsymbol{\beta}\rangle_{\mathcal{V}} \quad \dots \quad \langle \boldsymbol{\beta}_m, \boldsymbol{\beta}\rangle_{\mathcal{V}}]^{\mathsf{T}},$$

thus any $\boldsymbol{\beta} \in \mathcal{V}$ can be fully determined by the m inner products. Especially, if the basis is orthonormal, then

$$\begin{aligned} G_{\mathcal{V}} &= E_m, \\ \boldsymbol{y} &= [\langle \boldsymbol{\beta}_1, \boldsymbol{\beta}\rangle_{\mathcal{V}} \quad \langle \boldsymbol{\beta}_2, \boldsymbol{\beta}\rangle_{\mathcal{V}} \quad \dots \quad \langle \boldsymbol{\beta}_m, \boldsymbol{\beta}\rangle_{\mathcal{V}}]^{\mathsf{T}}. \end{aligned}$$

Consider then a linear transformation $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$. For any $\boldsymbol{\alpha} \in \mathcal{U}$, $\mathscr{A}(\boldsymbol{\alpha})$ can be uniquely determined by the m inner products $\langle \boldsymbol{\beta}_j, \mathscr{A}(\boldsymbol{\alpha})\rangle_{\mathcal{V}}$, $j = 1, 2, \dots, m$. The action of the linear transformation $\mathscr{A}$ on any $\boldsymbol{\alpha} \in \mathcal{U}$ can be fully determined by

$$\langle \boldsymbol{\beta}_1, \mathscr{A}(\boldsymbol{\alpha})\rangle_{\mathcal{V}}, \quad \langle \boldsymbol{\beta}_2, \mathscr{A}(\boldsymbol{\alpha})\rangle_{\mathcal{V}}, \quad \dots, \quad \langle \boldsymbol{\beta}_m, \mathscr{A}(\boldsymbol{\alpha})\rangle_{\mathcal{V}}. \tag{1}$$

4.5 Adjoint transformation

Since β_j, $j = 1, 2, \ldots, m$, are a basis of $\mathcal{V}$, when the data in (1) on the last slide are known,

$$\langle \beta\,,\, \mathscr{A}(\alpha)\rangle_{\mathcal{V}} \qquad (2)$$

can be also fully determined for any $\alpha \in \mathcal{U}$ and $\beta \in \mathcal{V}$. Thus the action of $\mathscr{A}$ can be fully determined by (2) when α ranges over $\mathcal{U}$ and β ranges over $\mathcal{V}$. Since $\mathcal{U}$ is also a unitary space, we may ask whether there exists $\mathscr{B} \in \mathcal{L}(\mathcal{V} \mapsto \mathcal{U})$, such that

$$\langle \mathscr{B}(\beta)\,,\, \alpha\rangle_{\mathcal{U}} = \langle \beta\,,\, \mathscr{A}(\alpha)\rangle_{\mathcal{V}}\,, \quad \forall \alpha \in \mathcal{U}\,, \quad \beta \in \mathcal{V}\,.$$

4.5 Adjoint transformation

- The existence and the construction of adjoint transformation

Theorem 4.20 Let $\boldsymbol{\alpha}_i$, $i = 1, 2, \ldots, n$, be an orthonormal basis of $\mathcal{U}$, $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$. $\mathscr{A}^*$: $\mathcal{V} \mapsto \mathcal{U}$ defined as

$$\mathscr{A}^*(\boldsymbol{\beta}) \triangleq \sum_{i=1}^{n} \langle \mathscr{A}(\boldsymbol{\alpha}_i), \boldsymbol{\beta} \rangle_{\mathcal{V}} \boldsymbol{\alpha}_i \qquad (1)$$

is the adjoint transformation of $\mathscr{A}$.

- Notice: when the basis of the U is orthonormal, the, the adjoint transformation $\mathcal{A}^*$ has a very simple form in relation to the linear transformation $\mathcal{A}$,

4.5 Adjoint transformation

•Indeed, when the bases are orthonormal, the matrix representation of the ajoint transformation $\mathscr{A}^*$ has a simple form in relation to the matrix representation of $\mathscr{A}$.

Theorem 4.21 Let $\alpha_1, \alpha_2, \ldots, \alpha_n$ and $\beta_1, \beta_2, \ldots, \beta_m$ be orthonormal bases of $\mathcal{U}$ and $\mathcal{V}$ respectively, the representation of $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$ with respect to these bases be A. Then the matrix representation of the adjoint transformation $\mathscr{A}^*$ with respect to the same bases is A^*.

4.5 Adjoint transformation

- Properties

Theorem 4.22 Let $\mathcal{U}$ and $\mathcal{V}$ be n- and m-dimensional unitary spaces respectively, $\mathscr{A} \in \mathcal{L}(\mathcal{U} \mapsto \mathcal{V})$, $\mathscr{A}^* \in \mathcal{L}(\mathcal{V} \mapsto \mathcal{U})$ be the adjoint transformation of $\mathscr{A}$, then

1. $\mathcal{N}(\mathscr{A}) \oplus \mathcal{R}(\mathscr{A}^*) = \mathcal{U}$;

2. $\mathcal{R}(\mathscr{A}) \oplus \mathcal{N}(\mathscr{A}^*) = \mathcal{V}$.

4.6 Normal transformation

Z. Song

- Background

•Suppose, a linear transformation on n-dimensional space, has n linearly independent eigenvectors,

•which implies if we select the n linearly independent eigenvectors as basis, the matrix representation is diagonal, (each Jordan form is diagonal))

•We call those linear transformations that can be similarly reduced to diagonal form –simple transformations

•The corresponding matrix representation is called simple matrices;

•Simple matrices are those can be diagonalized.

•In this section, we consider a special simple transformation (matrix), whose eigenvectors forms an orthonormal system, (orthormormal eigenbasis)--Normal transformation, the corresponding matrix representation is called normal matrices

4.6 Normal transformation

Z. Song

- Definition

Definition 4.17 A linear transformation $\mathscr{A}$ on a unitary space $\mathcal{V}$ is normal if it has an orthonormal eigenbasis. The matrix presentation A of a normal transformation is called a normal matrix.

- What does it like?

Theorem 4.23 $\mathscr{A}$ is normal if and only if $\mathscr{A}^* \circ \mathscr{A} = \mathscr{A} \circ \mathscr{A}^*$.

- Proof on pg 159, please read after class,

4.6 Normal transformation

Z. Song

- Sketch of the sufficient part that may help for your self-reading

- 1. Relations of eigenvalues and eigenvectors of $\mathscr{A}$, $\mathscr{A}^*$, $\mathscr{A} \circ \mathscr{A}^*$.
  - Eigenvectors are the same,

  if λ_j is an eigenvalue of $\mathscr{A}$, then $\bar{\lambda}_j$ is an eigenvalue of $\mathscr{A}^*$, and $|\lambda_j|^2$ is an eigenvalue of $\mathscr{A} \circ \mathscr{A}^*$.

- 2. $\mathscr{A} \circ \mathscr{A}^*$ is simple, and so are $\mathscr{A}$ and $\mathscr{A}^*$.

- 3. Eigenvectors can be an orthomormal system:

first show that the eigenvectors for distinct eigenvalues are orthogonal, for eigenvectors for the same eigenvalue can be orthonormalized via Gram-Schmidt method.

4.6 Normal transformation

- Three special normal transformations

Theorem 4.24 Self adjoint, skew self adjoint and unitary transformations are all normal.

Proof: Trival for the first two cases.
$\mathscr{U} \in \mathcal{L}(\mathcal{V})$ is unitary if $\langle \mathscr{U}(\boldsymbol{\alpha}), \mathscr{U}(\boldsymbol{\beta}) \rangle = \langle \boldsymbol{\alpha}, \boldsymbol{\beta} \rangle$ for all $\boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathcal{V}$. However, since

$$\langle \mathscr{U}(\boldsymbol{\alpha}), \mathscr{U}(\boldsymbol{\beta}) \rangle = \langle \boldsymbol{\alpha}, (\mathscr{U}^* \circ \mathscr{U})(\boldsymbol{\beta}) \rangle$$

for all $\boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathcal{V}$, we must have $\mathscr{U}^* \circ \mathscr{U} = \mathscr{E}$. In a similar manner we can show that $\mathscr{U} \circ \mathscr{U}^* = \mathscr{E}$. Thus $\mathscr{U}^* \circ \mathscr{U} = \mathscr{U} \circ \mathscr{U}^*$ and $\mathscr{U}$ is normal.

4.6 Normal transformation

- Properties of normal matrices

Theorem 4.25 Let $A \in \mathcal{C}^{n\times n}$, then A is normal if and only if there exists $U \in \mathcal{U}^{n\times n}$ such that

$$U^*AU = \text{diag}\{\lambda_1\ ,\ \lambda_2\ ,\ \ldots\ ,\ \lambda_n\}\ .$$

Definition 4.18 Let A, $B \in \mathcal{C}^{n\times n}$. A is said to be unitarily similar to B if there exists a $U \in \mathcal{U}^{n\times n}$ such that

$$U^*AU = U^{-1}AU = B\,.$$

Lemma 41. If A is normal, then any matrix unitarily similar to A is normal.

Theorem 4.28 A is a normal matrix if and only if $AA^* = A^*A$.

4.6 Normal transformation

- How far is a matrix from a normal matrix?

Theorem 4.26 (Schur Lemma) Any $n \times n$ matrix A is unitarily similar to an upper (a lower) triangular matrix.

Theorem 4.27 (Schur Inequality) Let $A = (a_{ij}) \in \mathcal{C}^{n \times n}$ with the eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$, then

$$\sum_{i=1}^{n} |\lambda_i|^2 \leq \sum_{i,j=1}^{n} |a_{ij}|^2$$

The equal sign holds if and only if A is normal.

Theorem 4.29 (Continuatioon of Schur Lemma-Theorem 4.26) Let $A \in \mathcal{C}_n^{n \times n}$ be nonsingular, then A can be uniquely factorized as

$$A = UR \qquad (1)$$

where $U \in \mathcal{U}^{n \times n}$, R is an upper triangular matrix with positive diagonal elements.

4.6 Normal transformation

- Properties of special normal matrices

Corollary 4.4 An upper triangular normal matrix A is diagonal. An upper triangular unitary matrix with positive diagonal elements is the identity matrix.

Theorem 4.30 Let $A \in \mathcal{C}^{n\times n}$ be a normal matrix, then

1. A is Hermitian if and only if all eigenvalues of A are real;
2. A is skew-Hermitian if and only if all eigenvalues of A are purely imaginary;
3. A is unitary if and only if $|\lambda_i(A)| = 1$ for all $i = 1, 2, \ldots, n$.

4.7 Hermitian matrices and Hermitian forms

Z. Song

- Hermitian matrices

Theorem 4.31 Let $A \in \mathcal{C}^{n\times n}$, then

1. A is Hermitian if and only if for any $\boldsymbol{x} \in \mathcal{C}^n$, $\boldsymbol{x}^* A \boldsymbol{x}$ is real;

2. A is Hermitian if and for any $S \in \mathcal{C}^{n\times n}$, $S^* A S$ is Hermitian.

4.7 Hermitian matrices and Hermitian forms

Z. Song

•Properties of Hermitian matrices

Hermitian matrix has the following properties:

1. A is Hermitian, then A^k is Hermitian for all $k \in \mathcal{Z}_{0,+}$;

2. A is Hermitian and invertible, then A^{-1} is Hermitian;

3. If A is Hermitian (skew-Hermitian), then iA is skew-Hermitian (Hermitian);

4. If A and B are both Hermitian, then for any real numbers k and$l$, $kA + lB$ is Hermitian;

5. For Hermitian matrices A and B, AB is Hermitian if and only if $AB = BA$.

4.7 Hermitian matrices and Hermitian forms

Z. Song

- Real-imaginary decomposition of a matrix using Hermitian matrices

A can be represented equally as

$$A = \frac{A + A^*}{2} + \frac{A - A^*}{2} = \frac{A + A^*}{2} + \mathrm{i}\frac{(A - A^*)}{2\mathrm{i}} \tag{1}$$

$A_{\mathrm{s}} \triangleq \frac{A+A^*}{2}$ is called the symmetric part and $A_{\mathrm{ss}} \triangleq \frac{A-A^*}{2}$ is called the skew-symmetric part of A.

Using (1), $\boldsymbol{x}^* A \boldsymbol{x}$ can be represented as

$$\boldsymbol{x}^* A \boldsymbol{x} = \boldsymbol{x}^* \frac{A + A^*}{2} \boldsymbol{x} + \mathrm{i}\boldsymbol{x}^* \frac{A - A^*}{2\mathrm{i}} \boldsymbol{x} \tag{2}$$

Since

$$\left(\frac{A - A^*}{2\mathrm{i}}\right)^* = \frac{A^* - A}{-2\mathrm{i}} = \frac{A - A^*}{2\mathrm{i}}$$

$\frac{A-A^*}{2\mathrm{i}}$ is Hermitian.

4.7 Hermitian matrices and Hermitian forms

Z. Song

- Hermitian forms:

Hermitian form is defined as

$$f(x_1, x_2, \ldots, x_n) = \sum_{i,j=1}^{n} a_{ij}\bar{x}_i x_j$$

with the coefficients satisfying the condition $a_{ij} = \bar{a}_{ji}$. By the definition, Hermitian form is a complex conjugate quadratic homogeneous function of the n complex variables x_1, x_2, ..., x_n. Let

$$\boldsymbol{x} = [x_1 \quad x_2 \quad \ldots \quad x_n]^{\top} \in \mathcal{C}^n$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \bar{a}_{12} & a_{22} & \cdots & \vdots \\ \vdots & \cdots & \cdots & a_{n-1,n} \\ \bar{a}_{1n} & \cdots & \bar{a}_{n-1,n} & a_{nn} \end{bmatrix}$$

then $A^* = A$, and

$$f(x_1, x_2, \ldots, x_n) = \boldsymbol{x}^* A \boldsymbol{x}. \qquad (1)$$

For notational simplicity, we also denote $f(x_1, x_2, \ldots, x_n)$ by $f(\boldsymbol{x})$.

4.7 Hermitian matrices and Hermitian forms

- Hermitian forms:

Theorem 4.32 A Hermitian form $f(\boldsymbol{x}) = \boldsymbol{x}^* A \boldsymbol{x}$ can be reduced to the canonical form

$$f(\boldsymbol{x}) = f(\boldsymbol{y}) = \lambda_1 y_1 \bar{y}_1 + \lambda_2 y_2 \bar{y}_2 + \cdots + \lambda_n y_n \bar{y}_n$$

via a unitary transformation $\boldsymbol{x} = C\boldsymbol{y}$, where $\lambda_j \in \mathcal{R}$, $j = 1, 2, \ldots, n$ are the eigenvalues of A. Furthermore, if the rank of $f(\boldsymbol{x})$ is r, then the canonical form is

$$f(\boldsymbol{x}) = f(\boldsymbol{y}) = \lambda_1 y_1 \bar{y}_1 + \lambda_2 y_2 \bar{y}_2 + \cdots + \lambda_r y_r \bar{y}_r \,.$$

4.8 Positive definite Hermitian forms

Definition of (semi-)positive definite

Definition 4.19 Given the Hermitian form

$$f(\boldsymbol{x}) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} \bar{x}_i x_j = \boldsymbol{x}^* A \boldsymbol{x} \,,$$

if for any $\boldsymbol{x} \neq \boldsymbol{0}$ $f(\boldsymbol{x}) > 0$ (≥ 0), then $f(\boldsymbol{x})$ is said to be positive (semi-) definite and so is the associated Hermitian matrix A.

4.8 Positive definite Hermitian forms

Z. Song

Properties of positive definiteness

Theorem 4.33 Given the Hermitian form $f(\boldsymbol{x}) = \boldsymbol{x}^* A \boldsymbol{x}$, the following statements are equivalent:

1. A is positive definite;
2. for any invertible $n \times n$ matrix P, P^*AP is positive definite;
3. A has n positive eigenvalues;
4. there exists $n \times n$ invertible matrix P, such that $P^*AP = E$;
5. there exists $n \times n$ invertible matrix Q, such that $A = Q^*Q$;
6. There exists a unique upper triangular matrix R with positive diagonal elements such that $A = R^*R$ (Cholesky factorization).

4.8 Positive definite Hermitian forms

Algebraic criterion for positive definiteness

Theorem 4.33 (Sylvester Criterion) An $n \times n$ Hermitian matrix A is positive definite if and only if the n successive leading principal minors are all positive, that is

$$a_{11} > 0, \qquad \det \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} > 0$$

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} > 0, \quad \ldots, \quad \det \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} > 0.$$

4.8 Positive definite Hermitian forms

Z. Song

Properties of semi-positive definiteness

Theorem 4.35 Let A be Hermitian, then the following statements are equivalent:

1. A is positive semi-definite;
2. For any $n \times n$ invertible matrix P, P^*AP is positive semi-definite;
3. A has n nonnegative eigenvalues;
4. there exists an $n \times n$ invertible P such that
$$P^*AP = \begin{bmatrix} E_r & 0 \\ 0 & 0 \end{bmatrix}$$
5. there exists an $n \times n$ matrix Q of rank r such that
$$A = Q^*Q$$

4.8 Positive definite Hermitian forms

Another important property of (semi)-positive definiteness

Theorem 4.36 Let A be positive (semi-) definite Hermitian matrix. Then there exists a unique positive (semi-) definite Hermitian matrix H such that $A = H^2$. Furthermore, any matrix commutative with A is also commutative with H.

4.8 Positive definite Hermitian forms

Examples

Example 4.13 Show that for positive definite Hermitian matrices A and B, the roots of $\det(\lambda B - A)$ are all positive numbers.